

Linear regression

Tengku Muhammad Hanis Bin Tengku Mokhtar, PhD

Department of Community Medicine
Faculty of Medicine
Universiti Sultan Zainal Abidin

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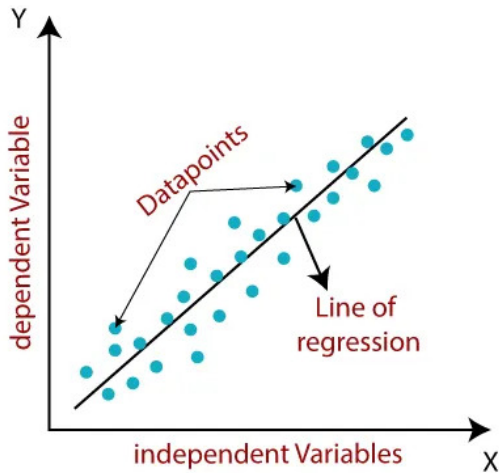
Multiple linear regression: Analysis

- Correlation tells us:
 - If the two variables are associated/related or not
 - How strong the association/relation between both variables
- Regression covers everything the correlation can tell us and a direction
 - If BMI increase by 1 unit, blood pressure will increase by 1.5 mmHg

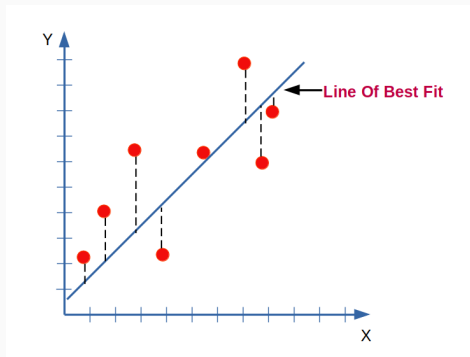
- Univariate vs. multivariate
 - Univariate - any statistical analysis that includes only 1 DV
 - Multivariate - any statistical analysis that includes > 1 DV
- Simple vs. multiple
 - Simple linear regression - univariable analysis (1 IV and 1 DV)
 - Multiple linear regression - multivariable analysis (> 1 IV and 1 DV)

Simple linear regression

- Aims: to describe the linear relationship between 1 numerical DV and 1 numerical/categorical IV
- When to conduct:
 - Random sample (theoretical assumption)
 - Numerical outcome
- Examples:
 - What is the effect of age on BMI?
 - What is the effect of family socio-economic status on exam score?



- The main idea of linear regression is to find the best fit line to describe the overall data
 - The line that minimise the difference between the observed and predicted value



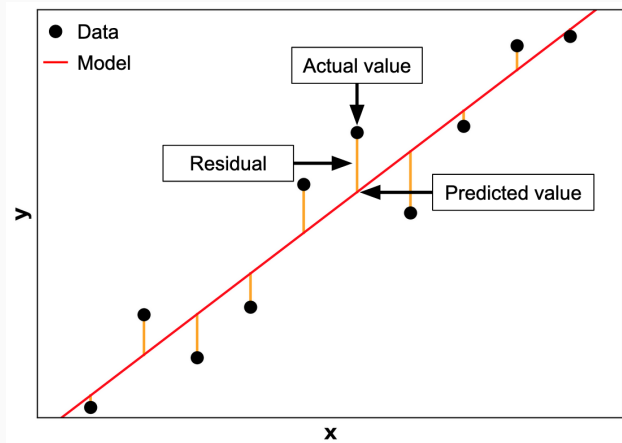
- Assumptions (LiNE):

1. Normality
2. Linearity
 - a. Overall linearity - residual vs. fitted values plot
 - b. Linearity of each continuous IV - residual vs. continuous IV plot
3. Equal variance - residual vs. fitted values plot

**Residual values: the difference between observed values and predicted values*

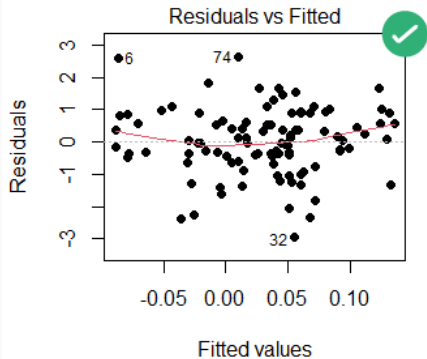
**Fitted values: predicted values predict by the regression model*

Predicted/fitted, actual, and residual value:

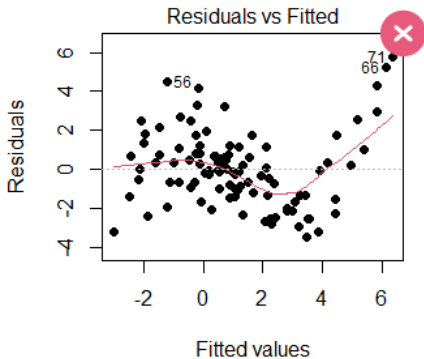


Linearity assumption:

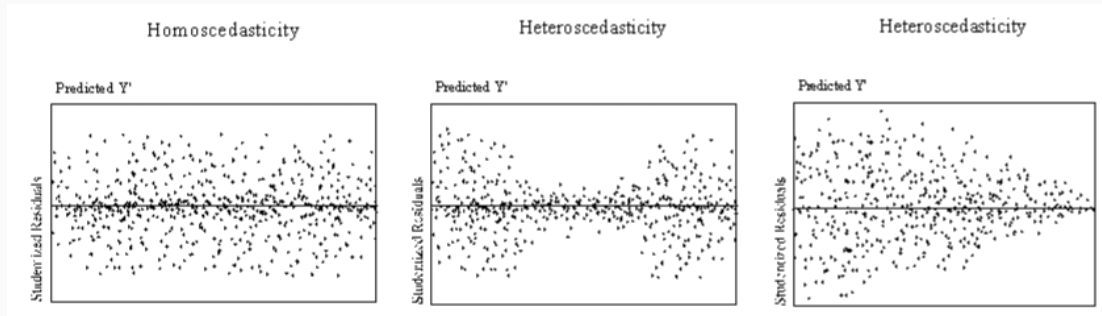
A: Linearity assumption satisfied:



B: Linearity assumption violated:



Equal variance:



Simple linear regression: Analysis

- Background: we want investigate the impact of width of the sepal on the length of the sepal among the iris flower
- Steps:
 1. State the hypothesis
 2. Check the assumptions
 3. Check the model fit
 4. Interpretation
 5. Conclusion

- Step 1:
 - H_0 : There is no relationship between width of the sepal and length of the sepal ($\beta_1 = 0$)
 - H_A : There is a relationship between width of the sepal and length of the sepal ($\beta_1 \neq 0$)

Packages

```
library(knitr)
```

```
library(ggplot2)
```

```
library(dplyr)
```

Load in the data

```
dat <- iris
```

```
# First 6 rows of the data
```

```
head(dat) %>% kable()
```

Sepal.Length	Sepal.Width	Petal.Length	Petal.Width	Species
5.1	3.5	1.4	0.2	setosa
4.9	3.0	1.4	0.2	setosa
4.7	3.2	1.3	0.2	setosa
4.6	3.1	1.5	0.2	setosa
5.0	3.6	1.4	0.2	setosa
5.4	3.9	1.7	0.4	setosa

```
# Run simple linear regression analysis
```

```
lr <- lm(Sepal.Length ~ Sepal.Width, data = dat)
```

- Step 2: Assumptions:

1. Normality

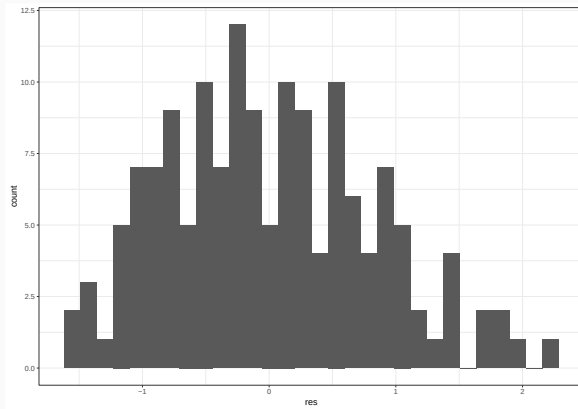
```
# Residual and fitted values
```

```
dat$res <- resid(lr)
```

```
dat$pred <- fitted.values(lr)
```

```
# Histogram
```

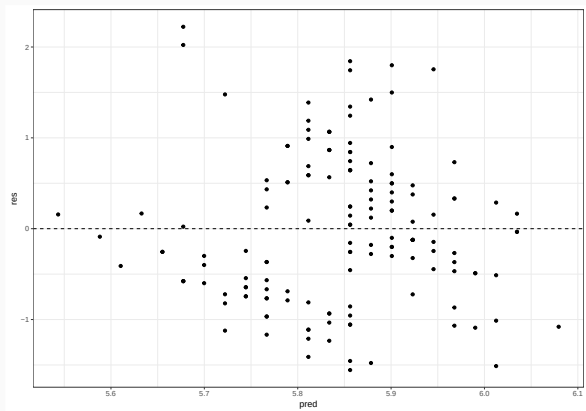
```
ggplot(dat, aes(res)) +  
  geom_histogram() +  
  theme_bw()
```



- Step 2: Assumptions:

2. Overall linearity
3. Equal variance

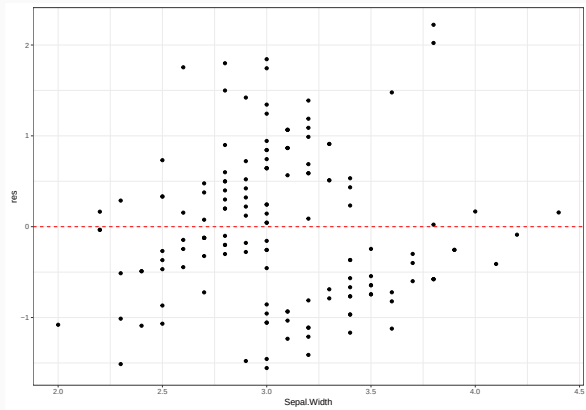
```
ggplot(dat, aes(pred, res)) +  
  geom_point() +  
  geom_hline(yintercept = 0, linetype = 2) +  
  theme_bw()
```



- Step 2: Assumptions:

- 4. Linearity of cont IVs

```
ggplot(dat, aes(Sepal.Width, res)) +  
  geom_point() +  
  geom_hline(yintercept = 0, linetype = 2, colour = "red") +  
  theme_bw()
```



- Step 3: Model fit:
 - Unadjusted R square:
 - How much the model/IVs can explain the the variation in DV
 - Always increase everytime IVs is added (significant or not)
 - So, the adjusted R square is proposed due to this limitation

- Step 3: Model fit:
 - Adjusted R^2 : The model explain 0.72% of the variance of the outcome

```
summary(lr)$adj.r.squared
```

```
[1] 0.007159294
```

- Step 4: Interpret:
 - P value = 0.152, which is > 0.05
 - 95% CI cross 0
 - There is no relationship between width of the sepal and length of the sepal
 - 1 cm increase in Sepal.Width will decrease 0.22 cm of Sepal.Length

P value

```
summary(lr)$coefficients
```

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	6.5262226	0.4788963	13.627631	6.469702e-28
Sepal.Width	-0.2233611	0.1550809	-1.440287	1.518983e-01

95% CI

`confint(lr)`

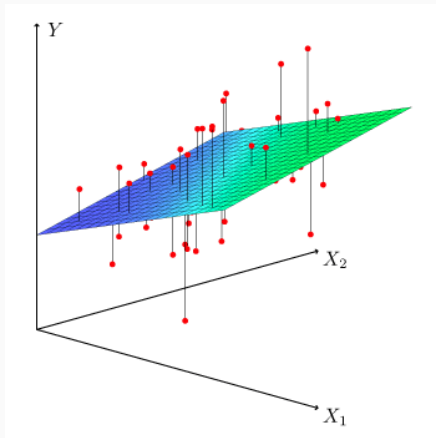
	2.5 %	97.5 %
(Intercept)	5.579865	7.47258038
Sepal.Width	-0.529820	0.08309785

- Step 5: Conclusion:
 - There is no relationship between width of the sepal and length of the sepal

Multiple linear regression

- Aims: to describe the linear relationship between 1 numerical DV and several numerical/categorical IVs
- When to conduct:
 - Random sample (theoretical assumption)
 - Numerical outcome
- Examples:
 - What is the effect of age and duration of exercise on BMI?
 - What is the effect of duration of study and family socio-economic status on exam score?

- In simple linear regression, we aim to find the best fit line
- In multiple linear regression, we aim to find the fit plane



- Assumptions (LiNE): similar to simple linear regression
 1. Normality
 2. Linearity
 - a. Overall linearity - residual vs. fitted values plot
 - b. Linearity of each continuous IV - residual vs. continuous IV plot
 3. Equal variance - residual vs. fitted values plot

Basic terms in regression

- Interaction:
 - When the effect IVs on the DV change according to the other IV
 - Example: effect of drug A on the blood pressure change according to age group
- Multicollinearity:
 - When two or more IVs in the regression model correlated with each other
 - Makes it difficult to account for the individual effect of the IVs on the DV
- Outliers - data points that are too different (too large or too small) from the rest of the data points

- Influential observations:
 - Outliers can be influential and can be not influential
 - Can be tested using Cook's distance

**Cook's distance describes how much your entire regression model would change if you deleted one specific data point*

Multiple linear regression: Analysis

- Background: we want investigate the effect of width of the sepal and species on the length of the sepal among iris flower
- Steps (we have extra steps):
 1. State the hypothesis
 2. Check for the interaction
 3. Check the assumptions
 4. Check the outliers and influential observations
 5. Check the multicollinearity
 6. Check the model fit
 7. Interpretation
 8. Conclusion

- Step 1:
 - H_0 : There is no relationship between width of the sepal, species, and length of the sepal ($\beta_1 = 0$)
 - H_A : There is a relationship between width of the sepal, species, and length of the sepal ($\beta_1 \neq 0$)

Packages

```
library(knitr)
```

```
library(ggplot2)
```

```
library(dplyr)
```

Load in the data

```
dat <- iris
```



```
# Run multiple linear regression analysis
```

```
mlr <- lm(Sepal.Length ~ Sepal.Width + Species, data = dat)
```

- Step 2: Interaction: p value > 0.05 (not significant)

```
int_mod <- update(mlr, .~. + Sepal.Width*Species)
```

```
summary(int_mod)$coefficients[5:6, ]
```

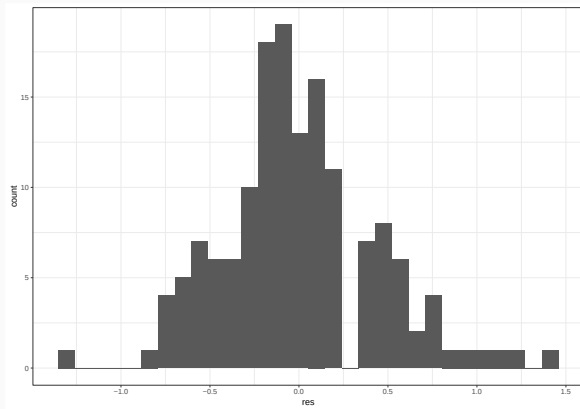
	Estimate	Std. Error	t value	Pr(> t)
Sepal.Width:Speciesversicolor	0.1745880	0.2598919	0.6717717	0.5028054
Sepal.Width:Speciesvirginica	0.2110448	0.2557557	0.8251811	0.4106337

- Step 3: Assumptions:

1. Normality

```
dat$res <- resid(mlr)
dat$pred <- fitted.values(mlr)

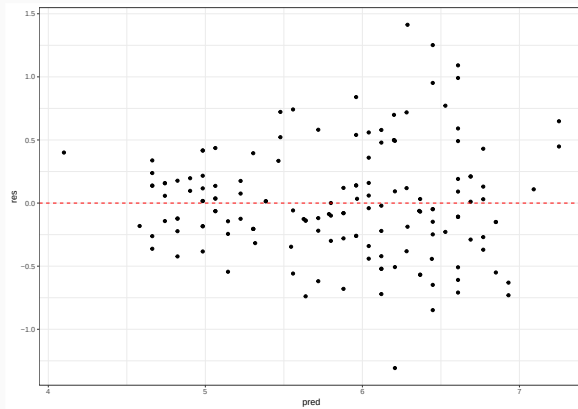
# Normality
dat %>%
  ggplot(aes(res)) +
  geom_histogram()
```



- Step 3: Assumptions:

2. Overall linearity
3. Equal variance

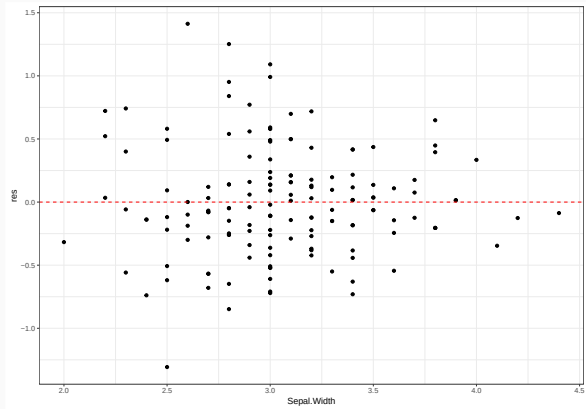
```
ggplot(dat, aes(pred, res)) +  
  geom_point() +  
  geom_hline(yintercept = 0, linetype = 2, colour = "red") +  
  theme_bw()
```



- Step 3: Assumptions:

- 4. Linearity of cont IVs

```
ggplot(dat, aes(Sepal.Width, res)) +  
  geom_point() +  
  geom_hline(yintercept = 0, linetype = 2, colour = "red")
```



- Step 4: Outliers and influential observations

```
out_data <- cooks.distance(mlr)
sum(out_data > 1)
```

```
[1] 0
```


- Step 5: Multicollinearity

```
car::vif(mlr) #all GVIF^(1/(2*Df)) should be < 3.16
```

	GVIF	Df	$GVIF^{1/(2 \cdot Df)}$
Sepal.Width	1.668844	1	1.291837
Species	1.668844	2	1.136590

- Step 6: Model fit:
 - Adjusted R^2 : The model explain 72.0% of the variance of the outcome

```
summary(mlr)$adj.r.squared
```

```
[1] 0.7202746
```

- Step 7: Interpret:
 - All p value < 0.05
 - All 95% CIs does not cross 0
 - There is a relationship between width of the sepal, species, and length of the sepal

95% CI

```
confint(mlr) %>% round(digits = 2)
```

	2.5 %	97.5 %
(Intercept)	1.52	2.98
Sepal.Width	0.59	1.01
Speciesversicolor	1.24	1.68
Speciesvirginica	1.75	2.14

- Step 7: Interpret:
 - 1 cm increase in Sepal.Width will decrease 0.8 cm of Sepal.Length after adjusted for the species
 - On average, the Sepal Length of the Versicolor species is 1.46 cm longer than the Setosa species, adjusting for sepal width
 - On average, the Sepal Length of the Virginica species is 1.95 cm longer than the Setosa species, adjusting for sepal width

P value

```
summary(mlr)$coefficients
```

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	2.2513932	0.3697543	6.088890	9.568102e-09
Sepal.Width	0.8035609	0.1063390	7.556598	4.187340e-12
Speciesversicolor	1.4587431	0.1121079	13.011954	3.478232e-26
Speciesvirginica	1.9468166	0.1000150	19.465255	2.094475e-42

- Step 8: Conclusion:
 - There is a relationship between width of the sepal, species, and length of the sepal